Table 1: **Characteristics of the analysed cohort, by diabetic peripheral neuropathy status.** Values are shown for all 187 analysed patients and separately for the 57 Unconfirmed and 130 Confirmed patients, for each of the 22 candidate predictors available to the models.

Characteristic	**Overall** (n=187)	**Unconfirmed** (n=57)	**Confirmed** (n=130)	**Effect size (95% CI)**	**p**	**q**
Profile						
SEX, male	62 (33.2)	15 (26.3)	47 (36.2)	OR 1.59 (0.80–3.16)	0.188	0.230
AGE, years	59 [50–66]	52 [39–63]	61 [53–67]	AUC 0.66 (0.57–0.74)	<0.001	0.001
SUBJ, yes	160 (85.6)	41 (71.9)	119 (91.5)	OR 4.22 (1.81–9.84)	<0.001	0.001
DM_DUR, years	5 [2–10]	2 [1–6]	7 [3–12]	AUC 0.68 (0.59–0.76)	<0.001	<0.001
INSULIN, yes ◇	78 (41.7)	17 (29.8)	61 (46.9)	OR 2.08 (1.07–4.04)	0.029	0.046
HBA1C, % ◇	7.5 [6.4–9.8]	7.2 [6.4–8.8]	7.6 [6.4–10.3]	AUC 0.55 (0.46–0.63)	0.327	0.379
Comorbidity						
HPN, yes ◇	112 (59.9)	33 (57.9)	79 (60.8)	OR 1.13 (0.60–2.12)	0.712	0.712
PAOD, yes ◇	8 (4.3)	1 (1.8)	7 (5.4)	OR 3.19 (0.38–26.52)	0.438	0.482
DSLPDMIA, yes ◇	97 (51.9)	24 (42.1)	73 (56.2)	OR 1.76 (0.94–3.31)	0.077	0.106
CKD, yes ◇	37 (19.8)	7 (12.3)	30 (23.1)	OR 2.14 (0.88–5.22)	0.088	0.114
GBS, yes	2 (1.1)	1 (1.8)	1 (0.8)	OR 0.43 (0.03–7.06)	0.518	0.543
Neurological examination						
DEC_VS, yes	96 (51.3)	11 (19.3)	85 (65.4)	OR 7.90 (3.73–16.73)	<0.001	<0.001
DEC_PPS, yes	92 (49.2)	15 (26.3)	77 (59.2)	OR 4.07 (2.05–8.07)	<0.001	<0.001
DEC_LTS, yes	90 (48.1)	17 (29.8)	73 (56.2)	OR 3.01 (1.55–5.86)	<0.001	0.002
DEC_AR, yes	94 (50.3)	9 (15.8)	85 (65.4)	OR 10.07 (4.53–22.38)	<0.001	<0.001
MNSI						
MNSI, score	5 [3–7]	4 [2–6]	6 [4–8]	AUC 0.67 (0.58–0.76)	<0.001	<0.001
Sudoscan						
FEET_MEAN_ESC, uS	53 [38–65]	60 [48–71]	48 [32–59]	AUC 0.32 (0.25–0.40)	<0.001	<0.001
FEET_PCT_ASYM, %	5 [2–11]	3 [1–6]	6 [3–12]	AUC 0.65 (0.56–0.73)	<0.001	0.002
HAND_MEAN_ESC, uS	58 [44–72]	63 [51–75]	56 [41–70]	AUC 0.39 (0.30–0.47)	0.013	0.022
HAND_PCT_ASYM, %	9 [4–14]	6 [4–11]	10 [4–17]	AUC 0.59 (0.50–0.67)	0.061	0.089
NS, score	53 [46–64]	61 [52–77]	51 [44–58]	AUC 0.28 (0.20–0.36)	<0.001	<0.001
CAS, %	30 [25–35]	28 [18–33]	31 [27–35]	AUC 0.64 (0.54–0.72)	0.003	0.006

Continuous variables are median [interquartile range] and were compared with the Mann–Whitney U test; binary variables are n (%) of the stated level and were compared with the chi-square test, or Fisher's exact test where any expected cell count was below 5. Effect size is the directional area under the receiver operating characteristic curve (AUC) for continuous variables and the odds ratio (OR) for binary variables; an AUC above 0.5 indicates higher values in the Confirmed group and an AUC below 0.5 indicates lower values. AUC confidence intervals are stratified percentile bootstrap intervals (2000 resamples). q denotes the Benjamini–Hochberg false discovery rate across the 22 candidate predictors. ◇ marks the features the counterfactual engine is permitted to vary.

Table 2: **Feature eligibility and target-leakage audit.** Role of each feature in the analysis together with its univariable discrimination, showing why nerve conduction variables are withheld from the models.

Feature	Role	AUC (95% CI)	q	Non-recordable, n (%)
Profile				
SEX	Predictor, fixed	0.549 (0.477–0.623)	0.230	–
AGE	Predictor, fixed	0.658 (0.568–0.745)	0.001	–
SUBJ	Predictor, fixed	0.598 (0.536–0.662)	0.001	–
DM_DUR	Predictor, fixed	0.682 (0.589–0.764)	<0.001	–
INSULIN	Predictor, actionable ◇	0.585 (0.507–0.655)	0.046	–
HBA1C	Predictor, actionable ◇	0.545 (0.459–0.629)	0.379	–
Comorbidity				
HPN	Predictor, actionable ◇	0.514 (0.435–0.587)	0.712	–
PAOD	Predictor, actionable ◇	0.518 (0.492–0.542)	0.482	–
DSLPDMIA	Predictor, actionable ◇	0.570 (0.495–0.647)	0.106	–
CKD	Predictor, actionable ◇	0.554 (0.500–0.607)	0.114	–
GBS	Predictor, fixed	0.495 (0.474–0.508)	0.543	–
Neurological examination				
DEC_VS	Predictor, fixed	0.730 (0.666–0.793)	<0.001	–
DEC_PPS	Predictor, fixed	0.665 (0.592–0.734)	<0.001	–
DEC_LTS	Predictor, fixed	0.632 (0.562–0.702)	0.002	–
DEC_AR	Predictor, fixed	0.748 (0.685–0.808)	<0.001	–
MNSI				
MNSI	Predictor, fixed	0.672 (0.585–0.756)	<0.001	–
Sudoscan				
FEET_MEAN_ESC	Predictor, fixed	0.321 (0.245–0.405)	<0.001	–
FEET_PCT_ASYM	Predictor, fixed	0.652 (0.564–0.733)	0.002	–
HAND_MEAN_ESC	Predictor, fixed	0.386 (0.299–0.473)	0.022	–
HAND_PCT_ASYM	Predictor, fixed	0.586 (0.499–0.673)	0.089	–
NS	Predictor, fixed	0.281 (0.196–0.364)	<0.001	–
CAS	Predictor, fixed	0.635 (0.545–0.725)	0.006	–
Nerve conduction study				
SSA_L	Excluded	0.073 (0.039–0.115)	<0.001	35 (18.7)
SSC_L	Excluded	0.226 (0.161–0.295)	<0.001	35 (18.7)
SPSA_L	Excluded	0.136 (0.087–0.195)	<0.001	53 (28.3)
SPSC_L	Excluded	0.205 (0.144–0.271)	<0.001	53 (28.3)
MCV_L	Excluded	0.183 (0.126–0.244)	<0.001	2 (1.1)
DL_L	Excluded	0.724 (0.652–0.798)	<0.001	2 (1.1)
CMAPANK_L	Excluded	0.080 (0.042–0.127)	<0.001	2 (1.1)
CMAPKNE_L	Excluded	0.075 (0.036–0.117)	<0.001	2 (1.1)
FWAVE_L	Excluded	0.779 (0.712–0.841)	<0.001	4 (2.1)
SSA_R	Excluded	0.081 (0.044–0.125)	<0.001	35 (18.7)
SSC_R	Excluded	0.230 (0.169–0.297)	<0.001	35 (18.7)
SPSA_R	Excluded	0.130 (0.081–0.186)	<0.001	53 (28.3)
SPSC_R	Excluded	0.217 (0.155–0.285)	<0.001	53 (28.3)
MCV_R	Excluded	0.149 (0.094–0.209)	<0.001	2 (1.1)
DL_R	Excluded	0.630 (0.541–0.713)	0.005	2 (1.1)
CMAPANK_R	Excluded	0.095 (0.052–0.142)	<0.001	2 (1.1)
CMAPKNE_R	Excluded	0.081 (0.039–0.130)	<0.001	2 (1.1)
FWAVE_R	Excluded	0.745 (0.674–0.812)	<0.001	8 (4.3)

AUC is the univariable directional area under the receiver operating characteristic curve for the feature used alone, with a stratified percentile bootstrap confidence interval. q-values are Benjamini–Hochberg false discovery rates computed within the 22 candidate predictors and, separately, within the 18 excluded nerve conduction variables. Non-recordable counts the patients whose nerve response was absent (recorded as *NR* or *NO F WAVE* in the source data and stored as zero); these zeros are retained in all summaries. ◇ marks the counterfactual-actionable features.

Table 3: **Data provenance and completeness.** Every alteration applied between the source spreadsheet and the analysed dataset.

Item	Change applied	Cells / records
Value recoding		
SPSA_L	*NR* → 0	53
SPSA_R	*NR* → 0	53
SPSC_L	*NR* → 0	53
SPSC_R	*NR* → 0	53
SSA_L	*NR* → 0	35
SSA_R	*NR* → 0	35
SSC_L	*NR* → 0	35
SSC_R	*NR* → 0	35
DM_DUR	*<1* → 1	11
FWAVE_R	*NR* → 0	7
FWAVE_L	*NR* → 0	3
CMAPANK_L	*NR* → 0	2
CMAPANK_R	*NR* → 0	2
CMAPKNE_L	*NR* → 0	2
CMAPKNE_R	*NR* → 0	2
DL_L	*NR* → 0	2
DL_R	*NR* → 0	2
MCV_L	*NR* → 0	2
MCV_R	*NR* → 0	2
DM_DUR	*>10* → 11	1
FWAVE_L	*NO F WAVE* → 0	1
FWAVE_R	*NO F WAVE* → 0	1
all binary features	*Y/M* → 1, *N/F* → 0	deterministic
Excluded records		
Patient 36	missing: NS, CAS	1
Patient 46	missing: DM_DUR, HBA1C	1
Patient 173	missing: NS	1

Generated from the audit trail written by the data loader, so the table cannot drift from the data actually analysed.

Table 4: **Nerve conduction study variables.** Descriptive summary of the variables withheld from model development because they enter the definition of the outcome.

Variable	Overall	**Unconfirmed** (n=57)	**Confirmed** (n=130)	Non-recordable	AUC (95% CI)	q
SSA_L, uV	8.9 [4.5–14.4]	17.9 [12.4–20.9]	6.6 [0.0–9.4]	0 (0.0) / 35 (26.9)	0.073 (0.039–0.115)	<0.001
SSC_L, m/s	46.7 [40.5–49.6]	49.6 [46.9–52.8]	43.8 [0.0–48.8]	0 (0.0) / 35 (26.9)	0.226 (0.161–0.295)	<0.001
SPSA_L, uV	6.0 [0.0–11.7]	13.2 [8.5–17.2]	3.5 [0.0–7.0]	1 (1.8) / 52 (40.0)	0.136 (0.087–0.195)	<0.001
SPSC_L, m/s	42.7 [0.0–49.1]	48.7 [43.1–52.6]	40.0 [0.0–45.9]	1 (1.8) / 52 (40.0)	0.205 (0.144–0.271)	<0.001
MCV_L, m/s	41.3 [37.5–44.9]	45.1 [42.1–47.0]	39.8 [36.2–43.0]	0 (0.0) / 2 (1.5)	0.183 (0.126–0.244)	<0.001
DL_L, ms	4.2 [3.6–4.7]	3.8 [3.4–4.1]	4.4 [3.8–4.9]	0 (0.0) / 2 (1.5)	0.724 (0.652–0.798)	<0.001
CMAPANK_L, mV	7.1 [4.1–10.7]	11.9 [10.1–15.1]	5.1 [2.8–7.7]	0 (0.0) / 2 (1.5)	0.080 (0.042–0.127)	<0.001
CMAPKNE_L, mV	5.2 [2.8–7.5]	8.7 [7.2–10.0]	3.5 [2.1–5.5]	0 (0.0) / 2 (1.5)	0.075 (0.036–0.117)	<0.001
FWAVE_L, ms	50.0 [46.2–54.7]	46.4 [43.4–49.2]	52.5 [48.3–56.2]	0 (0.0) / 4 (3.1)	0.779 (0.712–0.841)	<0.001
SSA_R, uV	8.8 [4.0–14.2]	17.6 [13.1–20.7]	6.1 [0.0–9.2]	0 (0.0) / 35 (26.9)	0.081 (0.044–0.125)	<0.001
SSC_R, m/s	45.5 [40.3–50.2]	49.6 [46.2–53.1]	43.1 [0.0–47.8]	0 (0.0) / 35 (26.9)	0.230 (0.169–0.297)	<0.001
SPSA_R, uV	6.3 [0.0–11.2]	12.6 [8.6–16.1]	4.3 [0.0–7.1]	1 (1.8) / 52 (40.0)	0.130 (0.081–0.186)	<0.001
SPSC_R, m/s	42.9 [0.0–49.1]	48.5 [44.1–51.8]	40.2 [0.0–45.9]	1 (1.8) / 52 (40.0)	0.217 (0.155–0.285)	<0.001
MCV_R, m/s	41.7 [38.3–45.2]	45.9 [43.7–48.6]	40.2 [36.1–42.3]	0 (0.0) / 2 (1.5)	0.149 (0.094–0.209)	<0.001
DL_R, ms	4.0 [3.5–4.5]	3.7 [3.4–4.2]	4.0 [3.6–4.6]	0 (0.0) / 2 (1.5)	0.630 (0.541–0.713)	0.005
CMAPANK_R, mV	7.5 [4.1–10.3]	12.2 [9.8–14.4]	6.0 [2.9–8.2]	0 (0.0) / 2 (1.5)	0.095 (0.052–0.142)	<0.001
CMAPKNE_R, mV	5.5 [2.7–7.3]	8.4 [7.1–9.8]	4.2 [2.0–5.9]	0 (0.0) / 2 (1.5)	0.081 (0.039–0.130)	<0.001
FWAVE_R, ms	49.9 [45.7–54.3]	46.1 [43.2–49.7]	51.7 [47.7–56.0]	0 (0.0) / 8 (6.2)	0.745 (0.674–0.812)	<0.001

Values are median [interquartile range]. Non-recordable gives the count and percentage of absent responses, Unconfirmed / Confirmed. These variables were withheld from model development; see Table 2.